METRIC

A-A-59173B
29 October 2024
SUPERSEDING
A-A-59173A
29 October 2009

COMMERCIAL ITEM DESCRIPTION

GREASE, SILICONE

The General Service Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers two types of silicone greases that are intended for application as a lubricant, a sealant, and as a preservative in missile systems.

2. CLASSIFICATION. The silicone grease shall conform to the following types and sizes:

2.1 Types.

Type I – Light consistency.
Type II – Medium consistency.

2.2 Sizes.

Size A – 8 ounce (oz) tube.
Size B – 10 pound (lb) can.
Size C – 5.3 oz tube.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DLA Aviation, VEB, 6090 Strathmore Rd, Richmond, VA 23237 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil

AMSC N/A FSC 9150

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3. SALIENT CHARACTERISTICS.

3.1 Material. The silicone grease shall have an unctuous (oily, soapy) consistency that is free of lumps, crusts, or granular particles.

3.2 Penetration. The grease shall have the following penetration when tested in accordance with ASTM D217, “Standard Test Methods for Cone Penetration of Lubricating Grease”: type I – penetration (tenths of a millimeter) range of 300-340; type II – penetration (tenths of a millimeter) range of 260-300.

3.3 Foreign particles. The grease shall have the following foreign particle requirements when tested in accordance with FED-STD-791, Method 3005, “Dirt Content of Grease”:

a. There shall not be more than 7500 particles that are of size 25 microns or greater per cubic centimeter in the grease.
b. There shall not be more than 1600 particles that are of size 75 microns or greater per cubic centimeter in the grease.
c. There shall not be any particles that are of size 125 microns or greater in the grease.

3.4 Oxidation stability. The oxygen pressure shall not decrease more than 34.47 kilopascals (kPa) during 50 hours of exposure of the material at a temperature of 150 ºC (302 ºF) when tested in accordance with ASTM D942, “Standard Test Method for Oxidation Stability of Lubricating Greases by the Oxygen Pressure Vessel Method”. An air bath may be substituted for the oil bath.

3.5 Dropping point. The grease shall have a dropping point greater than 190 ºC (374 ºF) when tested in accordance with ASTM D2265, “Standard Test Method for Dropping Point of Lubricating Grease Over Wide Temperature Range”.

3.6 Evaporation loss. The grease shall not lose greater than 3.0 percent of its original weight by evaporation when tested in accordance with ASTM D2595, “Standard Test Method for Evaporation Loss of Lubricating Greases Over Wide-Temperature Range”, at 150 °C (302 ºF).

3.7 Oil seperation. Type I silicone grease shall not lose greater than 5.0 percent of its original weight by bleeding. Type II silicone grease shall not lose greater than 4.0 percent of its original weight by bleeding. The percentage of bleeding of oil from the material shall be determined in accordance with ASTM D6184, “Standard Test Method for Oil Separation from Lubricating Grease (Conical Sieve Method)”, except that the heating of the entire test assembly will be maintained at 150 ºC (302 ºF) for a duration of 24 hours.

3.8 Solidification. The material shall show no evidence of solidification after exposure for 24 hours at a temperature of -55 ºC (-67 ºF). Evidence of solidification shall be tested in accordance with the following test procedure:

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- Put 20 millimeters of grease in a 2.54 centimeter (1 in) diameter test tube.
- Immerse the test tube in a suitable bath and cool to -55 ºC (-67 ºF) for a period of 24 hours.
- Remove the sample from the test tube and examine for evidence of solidification such as lumps, crusts, or granular particles.

4. REGULATORY REQUIREMENTS.

4.1 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and meeting the requirements of 29 Code of Federal Regulations (CFR) § 1910.1200, "Occupational Safety and Health Standards: Hazard Communication".

4.2 Recovered materials. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraphs 23.403 and 23.404(b) of the Federal Acquisition Regulation (FAR), as applicable. Feasibility of using recovered materials must be considered for silicone grease with respect to contamination purposes and meeting the salient characteristics of this commercial item description.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The following market acceptability criteria are necessary to document the quality of the product provided under this CID.

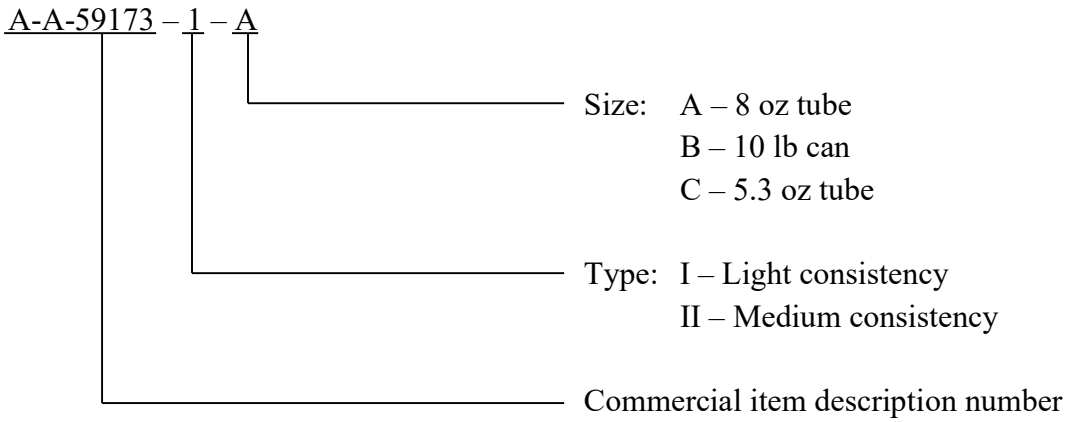
a. The company must have provided the silicone grease, meeting the salient characteristics of this CID, for at least 3 years.
b. The company must be able to show test data or lab results of meeting the salient characteristics of the silicone grease.
c. The company must provide a warranty of replacing the defective items.

6. PACKAGING. Preservation, packing and marking shall be as specified in the contract or order.

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7. NOTES.

7.1 Part identification number (PIN). The following part identification number procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Source of documents.

7.2.1 Electronic copies of CFR documents may be obtained from https://www.ecfr.gov.

7.2.2 Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/browse/index/far.

7.2.3 Electronic copies of federal specifications and standards may be obtained from https://quicksearch.dla.mil.

7.2.4 Electronic copies of ASTM documents may be obtained from https://www.astm.org.

7.3 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Part identification number (PIN).
c. Size of containers and quantity required.

7.4 National stock numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. This list may not be indicative of all possible NSNs associated with this document.

NSN	Type	Container size
9150-00-145-0161	I	8 oz tube
9150-01-228-3389	II	5.3 oz tube

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7.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27 Volumes 1 and 2. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

7.6 Key Words.

Lubricant
Missile systems
Preservative
Sealant

7.7 Changes from previous issues. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Reviewers:
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CIVIL AGENCY COORDINATING ACTIVITY:
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Preparing Activity:
DLA – GS3

(Project 9150-2024-006)

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